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Emerging investigator series: mercury mobility and methylmercury formation in a contaminated agricultural flood plain: influence of flooding and manure addition†

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The fate and the methylation of mercury (Hg) in the terrestrial environment are still poorly understood and although the main drivers of release and methylation of mercury in soils are known (low redox potential and microbial carbon availability) their interactions are not well understood. This is of concern since many agriculturally used floodplains, where the recurring flooding and agricultural practices (*e.g.* manure amendments) may have an impact on the fate and the biomethylation of Hg, are at the same time Hg-contaminated. In this study, we modified and validated existing methods to extract and analyze methylmercury (MeHg) by HPLC-ICP-MS in soils and we assessed the Hg and MeHg concentrations in three fields situated in a Hg polluted agricultural floodplain. Further, we incubated the top soil from the three studied fields for 11 days under flooded conditions in presence or absence of 2 mass% of cow manure, a common agricultural amendment in the area. Total Hg and MeHg concentrations ranged from <limit of detection (LOD, 0.012 mg kg^{-1}) to 28.2 mg kg^{-1} and from 1.2 to 7.8 μg kg^{-1} respectively. Hg was released to the soil solution after 12 hours with a maximum between day 2 and day 7. MeHg levels in the soil solution were <LOD although it was found in the soil before and after the incubation. The addition of cow manure to saturated soils led to an increase in the MeHg concentrations of the soil solid phase by up to fivefold to a maximum of 26.4 ± 1.1 μg kg^{-1} ($n = 3$). Our study demonstrates that the combination of low redox potential because of flooding with common agricultural practices such as the amendment of manures enhances the formation of toxic MeHg.

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Environmental significance

Mercury, a global pollutant, can be released into solution and transformed to the neurotoxic methylmercury in soils. However, few studies exist on the role of the drivers of methylmercury formation, including redox potential and organic matter concentrations. While the few existing studies focused on rice paddies and forest soils, we chose a floodplain site in Switzerland where industrial contamination meets agricultural use and where occasional flooding combines with increased organic matter concentrations because of the commonly applied animal manure. Our results from incubation experiments show that flooding leads to a release of Hg to the soil solution and that manure amendment leads to an increase of methylmercury formation.

Introduction

Sources of mercury (Hg) to the environment can be both natural (*e.g.*, volcanoes) and anthropogenic (*e.g.*, chemical industry, fossil fuel burning, incineration of waste or the extraction of metallic gold from alluvial washings).[1,2] The most important uses of Hg in the chemical industry are chloralkali and acetaldehyde production, where Hg serves as an electrode and a catalyst respectively.[3] Such activities can cause severe Hg pollution to the aquatic and terrestrial ecosystems with risks of mass poisoning, the Minamata catastrophe in Japan in the 1950s being a prime example.[4] There, the unintentional formation and release of monomethylmercury (MeHg), a potent neurotoxin, in the local marine bay and MeHg biomagnification in the food web due to its lipophilic nature,[5] caused the death of almost 1000 people. This was due to the high levels of MeHg in fish and shellfish combined with a fish-rich diet of the local people.[4,6,7] Because of the catastrophe in Minamata, research focused on Hg biomethylation, first in the aquatic environment and later in the terrestrial environment with a focus on paddy rice and forest soils.[8] However the drivers behind Hg behavior and its methylation in soils are not well understood.[3,9–11]

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In soils, Hg mobility and methylation are mainly governed by the redox potential (Eh), dissolved/soil organic matter (DOM/SOM), sulfur (S) and selenium (Se) concentrations, pH and both the presence of iron (Fe-) and manganese (Mn-)oxy-hydroxides.[12–15] MeHg is assumed to be mainly formed by sulfate-reducing bacteria (SRB), Fe-reducing bacteria (FeRB) and methanogens under anoxic conditions from dissolved inorganic mercury (Hg^{2+}).[9,16–18] Further, it is agreed, that Hg methylation depends on the speciation of both dissolved and solid-phase inorganic Hg^{2+} rather than the total Hg pool.[9,19,20] DOM plays a key role in Hg methylation and mobility, since on one hand it serves as a carbon (C) source and an electron donor for Hg methylators and it was shown to influence the Hg bioavailability by complexing Hg^{2+}.[11,21–23] The affinity of Hg to both DOM and SOM is well documented. In organic molecules, Hg shows the highest affinity to thiol functional groups but also carboxyl and amino groups are complexing Hg.[12,13,24,25] Recent work showed that also DOM quality (*e.g.* molecular weight or chemical composition originating from different sources) plays an important role in Hg bioavailability and methylation.[26–28] Experiments of Zhao and co-workers[23] showed that the presence of DOM inhibited Hg methylation by an Fe-reducing bacterium, but enhanced Hg methylation by a sulfate-reducing bacterium. In the latter study, it was suspected that thiol functional groups in DOM and on the cell's surfaces compete for Hg. Fe and Mn oxy-hydroxides are important sorbents of Hg and organic molecules. Consequently, organo–Hg complexes may sorb on the mineral surfaces and be subsequently released during reductive dissolution of the oxy-hydroxides.[29–31] Recent work showed that mineral bound OM-Hg and POM-Hg can act as a direct Hg source for Hg methylators.[32] Se also seems to impact methylation by limiting Hg availability due to the formation of Se–Hg complexes and mercuric selenide if selenide is present.[11] Fundamental questions about the biomethylation itself such as the mechanism of Hg uptake by methylating microorganisms and the biochemical pathway by which MeHg is produced still remain unanswered.[33]

Recurring reducing conditions and OM supply are the main methylation promoters in soils. For instance, ecosystem-scale studies reported higher MeHg in forest soils after flooding as a result of the change to anoxic conditions and increased DOM supply through decomposing vegetation.[34–36] Another experiment in rice paddies showed that flooding increased the Hg bioavailability and methylation,[37] while soil incubation experiments reported Hg mobilization and methylation in flooded soils.[38,39] The addition of organic amendments, such as rice straw, in paddy soils increased the methylation rate of Hg, likely because it provided additional DOM as C source for the methylating organisms[40,41] The application of mineral fertilizers can also increase the methylation rate of Hg in agricultural systems, because it enhances the microbial activity.[42] In forest soils, fertilizers seem to enhance methylation as a result of increasing pH and microbial activity.[43]

There is still a lack of experimental studies that investigate the influence of flooding and common agricultural practices, such as manure addition, on the behavior of Hg although agriculture on Hg-contaminated floodplain soils might present substantial risks to human health. Decreasing Eh in soils leads to Hg^{2+} mobilization into the soil solution as a result of the reductive dissolution of Mn- and Fe-oxyhydroxides and associated release of adsorbed or occluded Hg.[3,44–49] The few existing studies on the topic focused on the influence of water logging in forest soils[50,51] and flooding and OM concentrations in rice paddies,[40–42,52–54] while studies on temperate agricultural fields have been scarce,[38,55] although Hg-contaminated sites can often be found in floodplains.[56]

Here, we investigated the release and the methylation of Hg in agricultural floodplain soils contaminated by an acetaldehyde-producing company in order to better characterize the release and methylation of Hg and assess the potential future threat that Hg in agriculturally used Hg-contaminated floodplain soils represents. To do so, we modified and validated a fast extraction and analysis technique for MeHg in soils using HPLC-ICP-MS. Then, we determined the horizontal and vertical spatial distribution of Hg and MeHg in soil. Thirdly, we conducted an incubation experiment with soils from the same three fields to study the influence of flooding and manure amendment on Hg release and methylation.

Methods

Reagents

During the whole study, ultrapure water from a laboratory water system (MilliQ, ≥18.2 MΩ resistivity, Merck Millipore Inc., Burlington, MA, USA) was used for extractants and standard solutions, whereas a central system supplied deionized (DI) water for the incubation experiment and the various washing solutions. MeHg standard solutions were prepared by dissolving methylmercury(II)chloride (Sigma-Aldrich, Buchs, CH) in a small volume of methanol and then in ultrapure water. Multi-element standard (ICPMS-71A, Inorganic Ventures, Christianburg VA, US), and Hg standard (Fluka, Buchs, CH) were used for the total element analyses. Extractions were conducted with hydrochloric acid (HCl, 35%, Supra quality) from Roth (Karlsruhe, DE) and Dichloromethane (DCM, pro analysi) from Fisher Scientific (Loughborough, UK). The HPLC mobile phase (98% of 0.1% w/v L-cysteine + 0.1% L-cysteine·HCl·H_2O (pH = 2.3) and 2% methanol) was prepared with L-cysteine (Merck KGaA, Darmstadt, DE), L-cysteine·HCl·H_2O (VWR Chemicals, Leuven, BE) and HPLC grade methanol (Merck Millipore Inc., Burlington, MA, USA). Nitric acid (HNO_3, 69%, Supra quality) from Roth and hydrogen peroxide (H_2O_2, Suprapur quality) from Fluka were used for total digestions.

Study sites and soil sampling

We sampled soils (all Fluvisols[57]) in three fields from an agricultural floodplain contaminated with Hg originating from an acetaldehyde-producing company. The investigated fields are situated in Switzerland, in the Rhone Valley, between the cities of Visp and Raron. Between the 1930s and the 1970s, the company released Hg in a discharge canal linked to the Rhone river which then flows into the Lake Geneva. This canal was periodically cleaned, and the sediments were disseminated in

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the valley, including the surrounding agricultural fields.[58] The three sampling sites were situated along this canal which is called the Grossgrundkanal. They were chosen because they fulfilled the following conditions: they were (I) situated in the floodplain of the valley, (II) agriculturally used, and (III) contaminated with Hg (preliminary data obtained from the Agency for the Environment of the Canton Valais). All three soils have a silt loam texture, neutral to slightly alkaline pH, and a low to average organic carbon concentration in the topsoil (Table 1). At each site, soil cores of 50 cm depth were taken with a Pürckhauer corer (diameter 2 cm). Cores were drilled in triplicate at four distinct distances (5, 10, 20, 40 m) from the canal. For sampling, the plant cover was removed, and the core was divided into five depth intervals: 0–10, 10–20, 20–30, 30–40 and 40–50 cm. At each site, the three samples of the same depth and distance were pooled to result in a total of 60 composite samples. In addition to the soil cores, composite topsoil samples (0–10 cm, 1–2 kg) were collected at a distance of 5 m from the canal at each sampling site. These topsoil samples were used for the incubation experiment and kept fresh in a closed plastic bag (no drying) in the freezer at −20 °C prior to incubation. The soil core samples were first frozen at −20 °C just after sampling and then freeze-dried for approximately 48 h prior to extraction and analysis.[59] Dry samples were sieved to <2 mm, homogenized and ground manually with a ceramic mortar.

Soil characterization

For pH measurement, 10 g of soil was suspended in 25 mL of 0.01 M $CaCl_2$. Measurements were conducted 2 hours after equilibration. Soil moisture content was determined by weighing the soil before and after oven-drying and bulk density by collecting soil samples in the field in 100 cm^3 stainless steel rings. The loss on ignition (LOI) technique (550 °C for two hours muffle furnace) was used to determine the organic matter concentration. The C and N concentrations were measured before and after the measurement of LOI to determine total and inorganic C (and organic C as the difference between total and inorganic C) using an elemental analyzer (vario El cube, Elementar Analysensysteme, Langenselbold, Germany). DOC in soil solution was measured with a vario TOC cube (Elementar Analysensysteme, Langenselbold, Germany). Eh and pH were measured with an Eh electrode (SenTix ORP, WTW, Dinslaken, Germany) and a pH electrode (SenTix 21, WTW, Dinslaken, Germany) connected to a pH meter (pH 330i, WTW). For grain size analyses soil samples were first treated with H_2O_2 in order to remove organic matter and then dispersed in a solution of 22 mM sodium carbonate and 18 mM sodium hexametaphosphate. before the particle-size composition was measured on a MasterSizer 2000 (Malvern Panalytical Ltd., Malvern, UK). The concentration of sulfate (SO_4^{2-}) in soil solution was measured using ion chromatography (ICS-900, Dionex, Sunnyvale CA, USA).

Microwave-assisted digestion for total concentrations

Each sample from the soil cores as well as a representative aliquot of the topsoil used for incubation were digested using a microwave-assisted technique (Ethos contFlow 1600, milestone, Shelton CT, US) with 8 mL of 69% HNO_3 and 2 mL 30% H_2O_2 for 0.1 g of soil. For each of the three sites, two samples were digested in triplicates. For quality control, a Certified Reference Material (CRM, NIST San Joaquin Soil 2709a) and a blank sample were included in each microwave run. After digestion, each sample was brought to 20 mL with ultrapure water.

Soil extractions for MeHg

Typically, MeHg extraction and analysis was conducted on the soils before and after incubation and on a subset of samples from the cores (5 and 10 m distance to canal, 0–10, 10–20, 20–30 cm depth). Commonly, MeHg in soils and sediments only accounts for a small fraction of the total Hg concentration (<1%).[9] Hence, there was a need for a selective extraction for MeHg. Therefore, we adapted the method developed by Brombach and coworkers in order to make it compatible with HPLC-ICP-MS measurement.[60] Briefly, 50 mg of sample was suspended in 5 mL of 35% HCl and 5 mL ultrapure water in a 20 mL borosilicate glass vial (Wheaton, Milleville, NJ, UK). After 30 min of ultra-sonication, the vial was centrifuged for 3 min at 680 g (3500 rpm) and the supernatant transferred into a 50 mL separating funnel. Then, the lipophilic organic Hg was extracted two times by shaking with DCM in a separating funnel (2 × 4 mL). The two DCM solution thus obtained were combined in

Table 1 Soil properties of the investigated fields (mean ± sd). Soil properties refer to the average of 5 depth layers (10 cm layers, 0–50 cm) if not mentioned otherwise

	Site A	Site B	Site C
Grown crop	Maize (harvested)	Wheat (harvested)	Grass (cut)
GPS coordinates	46.30210N 7.81182E	46.30089N 7.81949E	46.30040N 7.82237E
pH	7.4–8.1	7.6–8.0	7.3–7.8
Texture:	Silt loam	Silt loam	Silt loam
Sand (%, $n = 9$)	18.0 ± 1.5	23.1 ± 7.3	26.4 ± 6.2
Silt (%, $n = 9$)	69.1 ± 1.5	64.5 ± 4.4	64.8 ± 4.2
Clay (%, $n = 9$)	12.9 ± 1.5	12.4 ± 3.3	8.9 ± 2.2
Moisture content (%, $n = 20$)	25.6 ± 1.1	25.6 ± 1.8	27.0 ± 1.8
Bulk density (g cm^{-3}, $n = 3$)	1.2 ± 0.1	1.5 ± 0.1	1.3 ± 0.1
Organic C concentration (mass%, 0–20 cm)	1.9 ± 0.2 ($n = 7$)	1.7 ± 0.2 ($n = 8$)	2.0 ± 0.3 ($n = 8$)

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a new 20 mL borosilicate glass vial. To prevent contamination, the separating funnels were washed by shaking first with a 10% HNO_3 solution, and then a 10% HCl solution, and DI water after each acid wash. The main modification we made to the Brombach *et al.*[60] method is that we used the HPLC mobile phase (L-cysteine solution) for the back extraction in order to make it compatible with a HPLC-ICP-MS measurement. In detail, 2 mL of the HPLC mobile phase were added to the DCM extract and the DCM was evaporated with a constant flow of N_2 by using a FlexiVap™ at 50 °C (GlasCol®, Terre Haute, IN, USA). The remaining L-cysteine solution was then weighed to determine its exact volume. The extracts were stored at 4 °C and analyzed within 48 hours. The method was tested using a CRM (ERM-CC580, estuarine sediment) and blank extractions with only the reagents and no soil sample.

Incubation experiment

Two days before the start of the experiment, the soil samples were thawed in a fridge at 4 °C and moist sieved <2 mm. The three prepared soils were then incubated in triplicate (80 g each) in 9 Erlenmeyer flasks (250 mL). Another 9 identical samples were incubated with 2 mass% additional OM (78.4 g soil and 1.6 g OM). The added OM was commercially available cow dung manure, which was dry and finely chopped (Hauert HBG Dünger AG, Grossaffoltern, Switzerland CH). The manure's Hg content was analyzed prior to the experiment (18.4 ± 6.9 μg kg^{-1}). All 18 Erlenmeyer flasks were then flooded with 120 mL of DI water. Each Erlenmeyer was equipped with a rhizon sampler connected to a syringe needle (Rhizon flex 5 cm, 0.15 μm pore size, Rhizosphere Research Products, Wageningen, NL) in order to sample soil solution and was then covered with parafilm. These microcosms were placed in a growth chamber (Binder, Tuttlingen, DE) at a constant temperature of 18 °C (mean summer temperature of the study site) and a relative air humidity of 60%. The incubation experiment lasted for 11 days.

The soil solutions were sampled after 6 h, 12 h, 1, 2, 3, 5, 7, 9 and 11 days. Samples were retrieved by applying a vacuum on the syringe needle in the form of a 5 mL vacutainer (Plastipak, Becton Dickinson, Franklin Lakes, NJ, USA). Since the soil solution passed through the rhizon sampler (pore size = 0.15 μm), no further filtering was necessary. At each sampling point, four different subsamples were taken (1: for total Hg and multi-element measurement, diluted in 1% HNO_3 and 0.5% HCl; 2: for Hg speciation diluted in the HPLC mobile phase; 3: for dissolved organic carbon (DOC) and dissolved organic nitrogen (DON) measurements diluted in ultrapure water, 4: for ion chromatography IC). Eh and pH values were measured immediately by using a 5th aliquot of the soil solution. All samples were stored in a fridge at 4 °C prior to analyses.

Measurement of metal and MeHg concentrations

Total metal concentrations (Hg, Fe, Mn) in soil digests and soil solutions were determined using Inductively-Coupled Plasma Mass Spectrometry (ICP-MS 7700× ICP-MS, Agilent Technologies, Waldbronn, Germany). The instrument's settings are shown in Table S1.† Indium served as an internal standard (*m*/*z* 115) and was continuously mixed with the sample through a T-piece. Hg was measured at *m*/*z* 201. Details about the special washout for Hg to avoid a memory effect can be found in Table S2.† Hg standards and samples were diluted with a mixture of 1% HNO_3 and 0.5% HCl. Standards and samples for all other elements were diluted in 1% HNO_3.

MeHg concentrations in soil extracts and soil solutions were determined by connecting a High-Pressure Liquid Chromatograph (HPLC 1200 Series, Agilent Technologies, Santa Clara, CA, USA) to the ICP-MS. The details of the method, initially suggested by Hight & Cheng[61] and optimized by Sannac & Chen,[62] are presented in Table S1.† Briefly, the mobile phase (98% of 0.1% w/v L-cysteine + 0.1% L-cysteine·HCl·H_2O (pH = 2.3) and 2% methanol) was set to a flow rate of 1 mL min^{-1}. The calibration standards for MeHg measurements were prepared daily by diluting the MeHg stock solution with the HPLC mobile phase.

Results and discussion

MeHg extraction and analysis method

Combining the modified MeHg extraction method of Brombach *et al.*[60] with the HPLC method based on Hight & Cheng[61] led to satisfying results. The recovery for MeHg in the CRM (ERM-CC580: 75 ± 4 μg kg^{-1} MeHg) was 102 ± 5% (n = 11). The LOD for MeHg was 1 μg kg^{-1} for soil or sediment samples and 10 ng L^{-1} for water samples (no extraction step). This LOD is slightly lower than the ones of other comparable methods listed in the review by Jagtap & Maher.[63] To further validate the method, a batch of homogeneous soil samples from our study area was split in two identical sets. One set was extracted in our laboratory and the other was sent to a commercial laboratory (Brooks Applied Labs, Bothell, WA, USA) which used the US EPA 1630 method with Ethylation, Purge & Trap, GC-Pyrolysis-CVAFS.[55] Fig. 1 illustrates a generally good match of the two methods. However, a paired t-test ($p < 0.001$) confirmed slightly higher concentrations with the method used in our laboratory. This can be explained by the fact that our recovery was consistently in the upper range compared to the recovery obtained by spiking soils using the US EPA 1630 method (102 ± 5% for CRM using our method *vs.* 77–106% for spiked samples using US EPA 1630). However, the Intraclass Correlation Coefficient (ICC): 0.9562, showed that the method we used is in good agreement with the conventional method for samples in this concentration range. The modified method represents an alternative for laboratories that are not equipped with specific analyzers since it uses HPLC-ICP-MS (ICP-MS can even be replaced by the cheaper AFS). Finally, although the LOD is higher for this method, 16 samples a day can be extracted and the run time for each sample is 4 min. This makes it interesting for risk assessment of large areas where many samples should be processed and thus opens the way to high throughput analysis of MeHg in soils.

Hg and MeHg concentrations in soil

Hg concentrations (total Hg if not mentioned otherwise) in the 60 soil samples showed values from below the LOD (0.012 mg kg^{-1})

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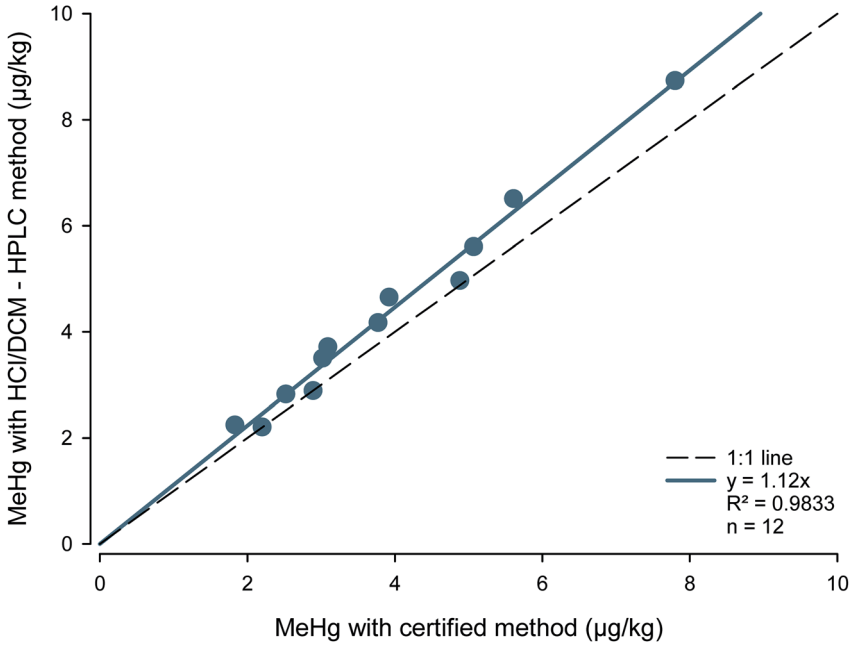


Fig. 1 Comparison of MeHg concentrations obtained from our modified extraction coupled to HPLC-ICP-MS measurement method with the certified method US EPA 1630.

up to 28.2 mg kg^{-1}. All three sites showed the highest Hg concentrations in the top 30 cm and closest to the canal. Two gradients of decreasing Hg concentration were prominent: with increasing (i) distance from the canal and (ii) soil depth (Fig. 2). These two gradients confirmed the reported past activities, when dredged sediments from the canal were deposited at its shore on the fields before further distribution for soil improvement or leveling soils in the surroundings.[58] Agricultural activities such as ploughing might have carried highly contaminated material from the shore further away from the canal and deeper into the soils.[58] The concentrations were all above the Swiss Remediation Threshold value of 2 mg kg^{-1} between 0 and 40 cm up to 10 m from the canal. For site B at 20 m distance from the canal, the concentrations exceeded this value also down to 40 cm while at site A only down to 20 cm. No samples exceeded this value at 40 m from the canal. Only few samples had Hg concentrations below the Swiss Investigation Threshold value of 0.5 mg kg^{-1} and these samples originated from below 40 cm and at least 10 m

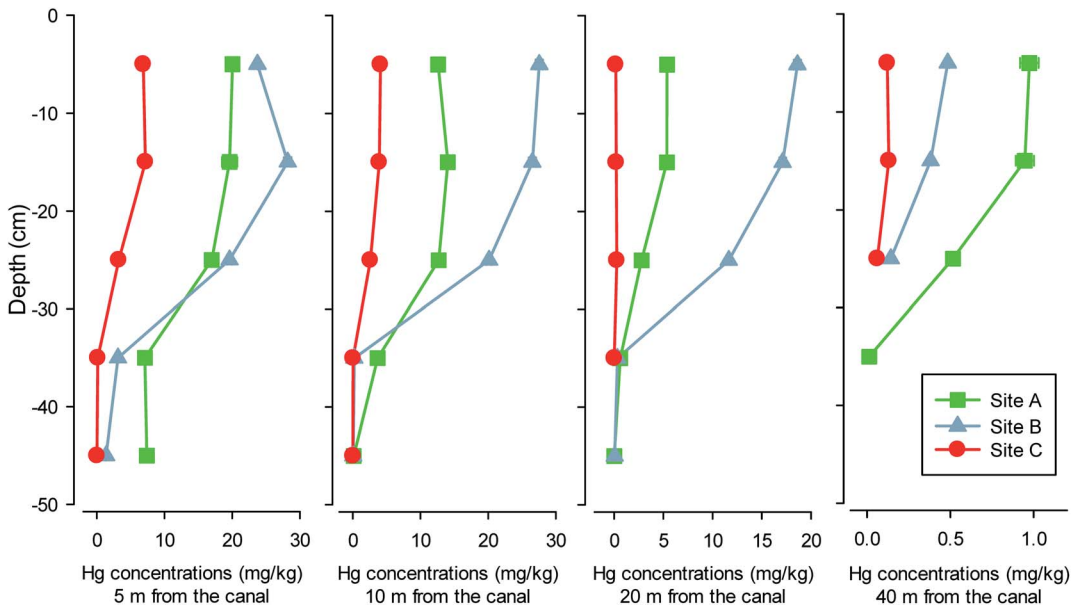


Fig. 2 Vertical distribution of total Hg concentrations in soils at three sites in the Valais, Switzerland (mean ± sd, $n = 3$).

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away from the canal, except for site C which exhibited lower concentrations than the other two sites.

Correlation analyses per site, distance, and depth between Hg concentrations and the percentage of the clay fraction showed positive relationships (Table 2, Fig. S1†). The clay fraction became less important for the Hg concentrations in soil with distance and depth. Furthermore, a one-way ANOVA showed that site C had a significantly lower clay ($p < 0.001$) and a significantly lower Hg concentration ($p < 0.05$) than sites A and B. In soils, Hg concentration and the clay grain size fraction are usually linked in soils depleted in organic matter, since Hg adsorption increases with smaller particle size (clay mineral content) and increasing specific surface area.[64] The weaker relationship between clay and Hg concentrations in the less Hg-contaminated samples can be explained by the mixing of strongly and little Hg-loaded clay originating from the dredged sediments and the background soil, respectively. We suggest two possible explanations for the lower Hg concentrations at site C. (i) It is known that the canal sediments have a silty clay texture while natural soils in the study area consist rather of silty sand[65] indicating that probably less sediment from the canal was amended to site C. (ii) The soil at site C contained less clay and thus fewer sorption sites for Hg and therefore some of the initial Hg input was leached or volatilized.

We measured MeHg concentrations in the soil samples with the highest Hg concentrations at each site, *i.e.* next to the canal (5 m) and in the plough horizon (0–30 cm). The 18 selected samples showed MeHg concentrations from 1.2 to 7.8 μg kg^{-1} (Fig. 3). It is interesting to note that site C had lower Hg but not lower MeHg concentrations than sites A and B. Consequently, site C showed the highest MeHg contribution to total Hg with a percentage of up to 0.084% (Table 3). Our results demonstrate that the MeHg concentration does not depend on total Hg concentrations in soils. We speculate instead that the MeHg/Hg concentration ratios might be related with bioavailable Hg concentrations and the composition and activity of the microbial community. The MeHg concentrations as well as the MeHg/Hg ratios in our study soils were in the same range as reported by Gilli *et al.*[55] in the same region, at a chlor-alkali industrial site in Italy by Cattani *et al.*,[66] or in the floodplain of a former Hg mining area in China by Wang *et al.*[37] However, Skyllberg *et al.*[67] reported a MeHg contribution to the total Hg concentrations of 17% in unpolluted boreal peat soil. High MeHg/Hg ratios might only be possible if total Hg in the soil is at a natural low level and soil conditions favor methylation.

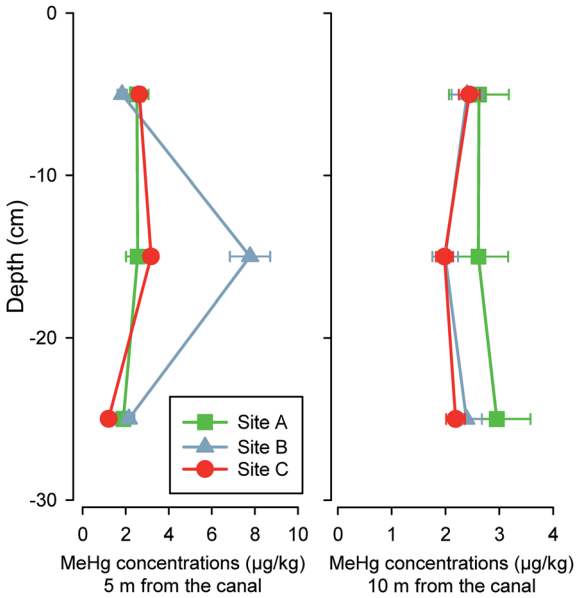


Fig. 3 Vertical distribution of MeHg concentrations in soils at three sites in the Valais, Switzerland (mean ± sd, $n = 3$).

Table 2 Correlation coefficients and error probabilities of clay content (%) with Hg concentration (mg kg^{-1}). n.s. is not significant

	p	Pearson correlation coefficient R
Site		
A	n.s.	0.61
B	<0.01	0.85
C	n.s.	0.37
Distance		
5 m	<0.001	0.92
10 m	<0.01	0.85
20 m	0.05	0.66
Depth		
0 cm	<0.05	0.73
20 cm	<0.05	0.77
40 cm	n.s.	0.53

Table 3 Concentrations of total Hg and MeHg and the MeHg/Hg ratio for the soil cores at 5 and 10 m distance from the canal and down to 30 cm (mean ± sd). The first number in the sample name denotes the distance from the canal (m), the second number soil depth (cm)

Sample	Hg [mg kg^{-1}]	MeHg [μg kg^{-1}]	MeHg/Hg [%]
5 m distance			
A5–0	20.0 ± 0.7	2.5 ± 0.5	0.013
A5–10	19.6 ± 1.1	2.6 ± 0.5	0.013
A5–20	17.0 ± 1.0	1.9 ± 0.4	0.011
B5–0	23.7 ± 0.5	1.8 ± 0.2	0.008
B5–10	28.2 ± 0.6	7.8 ± 0.9	0.028
B5–20	19.6 ± 0.4	2.1 ± 0.3	0.011
C5–0	6.8 ± 0.3	2.6 ± 0.2	0.039
C5–10	7.2 ± 0.3	3.2 ± 0.3	0.044
C5–20	3.2 ± 0.1	1.2 ± 0.1	0.037
10 m distance			
A10–0	12.6 ± 0.7	2.6 ± 0.6	0.021
A10–10	14.0 ± 0.8	2.6 ± 0.6	0.019
A10–20	12.7 ± 0.7	3.0 ± 0.6	0.023
B10–0	27.5 ± 0.6	2.4 ± 0.3	0.009
B10–10	26.5 ± 0.5	2.0 ± 0.2	0.008
B10–20	20.1 ± 0.4	2.4 ± 0.3	0.012
C10–0	4.1 ± 0.2	2.4 ± 0.2	0.060
C10–10	3.9 ± 0.2	2.0 ± 0.2	0.051
C10–20	2.6 ± 0.1	2.2 ± 0.2	0.084

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Incubation: Hg release into soil solution

The soils used for the incubation experiment had initial Hg concentrations at sites A, B and C of 28.0 ± 1.0, 26.7 ± 0.4 and 5.2 ± 0.1 mg kg^{-1}, respectively. The Hg release into the soil solution of the microcosms differed slightly between site A and the sites B and C showing a similar Hg release. Overall, there was an initial increase in dissolved Hg up to 30 to 40 μg L^{-1} within 2–5 days for microcosms of sites B and C and within 7 days for microcosm of site A with OM. In all microcosms, the Hg release to the soil solution took place earlier and was more pronounced when OM was added (Fig. 4a and b). However, soils of site A released only Hg if treated with OM. Also, Eh of site A remained higher in the second half of the incubation (Fig. S2c and d†), which was also accompanied by a smaller release of Fe and Mn into the soil solution (Fig. 4d and e). Apart from the incubation of site A without OM, the Hg concentrations in the soil solution increased until around day 3 in every microcosm. After that, the Hg concentration decreased to background values (after 6 h). This pattern, as well as the amount of released Hg from soil to soil solution (<1%) are consistent with other microcosm studies.[39,49,55,68,69] Of interest is the similar maximum amount of Hg being released to the soil solution in all OM-containing microcosms (site A: 38.9 μg L^{-1}, site B: 33.8 μg L^{-1} and site C: 23.1 μg L^{-1}, Fig. 4) although the soil from site C contained five times less Hg than the other two sites. The initial increase of dissolved Hg in the soil solution may be attributed to the dissolution of weakly bound Hg that is mobilized through mechanisms such as desorption from phyllosilicate and Fe and Mn (oxyhydr)oxide surfaces, dispersion of colloids <0.15 μm, cation exchange or mobilization as organo–Hg complexes by dissolved organic matter.[13,24,25,68] Furthermore, it can be the result of the reductive dissolution of Mn- and Fe-(oxyhydr) oxides-bound Hg.[47] Indeed, our data shows that in every microcosm Mn- and Fe-(oxyhydr)oxides reduction occurred, as reflected by the increase in total dissolved Fe and Mn concentrations (Fig. 4c–f). These two redox-sensitive metals are theoretically reduced at potentials of 450 mV for Mn and at 150 mV for Fe, which we did not measure for the bulk material, because the used Pt electrode detects an average Eh value of a larger soil volume, but which are likely reached at some microsites. Interestingly, the courses of Mn concentrations, but not Fe, in the soil solution of microcosms from sites B and C (with and without OM addition) correlated well with the increase of Hg concentrations in soil solution (Fig. 4a–d and S3†). Grigg *et al.*[70] observed a correlation between Mn and Hg concentrations during sequential extraction of contaminated soils in the region suggesting that a fraction of soil Hg was associated with Mn (oxyhdr)oxides. It is therefore likely that in this soil, Mn-(oxyhydr)oxide dissolution governed the Hg release, thus making Mn-(oxyhydr)oxide the primary binding site for Hg and a Hg sink under oxic conditions. This is further confirmed by the microcosms from site A without OM addition, which have no notable Hg release and constantly lower Mn concentrations than the soil solutions from sites B and C. Fe release consistently took place after the Hg concentrations started to diminish (Fig. 4e and f). It is therefore unlikely that Fe-(oxyhydr)oxide dissolution plays an essential role for the Hg release in the studied floodplain soil. Although the reductive dissolution of Mn-(oxyhydr)oxides seemed to play an important role for the Hg release into soil solution, it is not possible to completely exclude a role of organo-complexation of Hg by dissolved organic matter, since we observed a strong release of DOC into soil solution in the first days of the incubation (Fig. 4g and h).

Other studies also observed a decrease of Hg concentrations in the soil solution after reaching a maximum.[49,55,68] However, the exact mechanisms which took place in our experiment are difficult to elucidate. Possible explanations are (1) re-adsorption of Hg to clay minerals after the latter had lost some of their DOC cover, (2) Hg co-precipitation with sulfide nanoparticle aggregates[49] which may not have passed our rhizon sampler (pore size of 0.15 μm) or (3) the adsorption of Hg^{2+} onto the SOM functional groups (thiol, carboxyl and amino groups) which are well known to sorb Hg^{2+}.[12,13,24,25]

Dissolved Hg in the soil solution is potentially available for microbes which can methylate Hg to MeHg or reduce Hg^{2+} to Hg^0. The latter mechanism is a common detoxification mechanism of bacteria in such environments.[71,72] Since Hg^0 is volatile, it may have exited the microcosm. Frossard *et al.*[73] showed the presence of the bacterial detoxification genes (merA gene) in soils situated at a distance of about 150 meters from our study sites However, it is unlikely that sufficient Hg^{2+} was reduced to Hg^0 to solely explain the decrease in dissolved Hg in our microcosms.

Incubation: MeHg formation in the soil

MeHg was not detectable in the soil solution throughout the incubation period of 11 days during which the soil was kept flooded (LOD: 10 ng L^{-1}). This is to be expected since the calculated concentrations for MeHg in soil solution are mainly below LOD when using common soil-water partitioning coefficients ($\log(K_{d_{MeHg}}) = [1.3, 3.8]$).[74] MeHg was, however, measured in the soil before and after the incubation (Fig. 5). Before incubation, the microcosms of site A, B and C had MeHg concentrations of 3.5 ± 0.3, 6.1 ± 0.5 and 4.3 ± 0.4 μg kg^{-1} respectively. After the experiment, concentrations increased up to 26.4 ± 1.1 μg kg^{-1} at site B and 21.6 ± 2.2 μg kg^{-1} at site C which corresponded to a fivefold increase. While there were only minor changes in microcosms without OM treatment, the increase in the MeHg concentrations in microcosms with 2% OM was significant (*Z*-test: $p < 0.001$). In the microcosms with 2% OM, the MeHg contribution to total Hg concentrations increased during the experiment from 0.04% to 0.2%. The maximum increase of this percentage from 0.08% to 0.49% occurred in one microcosm from site C. In sediments, the threshold of 1% is rarely exceeded due to the cyclical nature of the methylation process (methylation and demethylation) and the often-observed inverse relationship of MeHg production with Hg concentrations.[9] However, there are some reported exceptions for unpolluted soils. Indeed, Skyllberg *et al.*[67] found MeHg contribution to the total Hg concentrations of between 1.2% and 17.2% in peaty stream bank soils of a pristine boreal forest.

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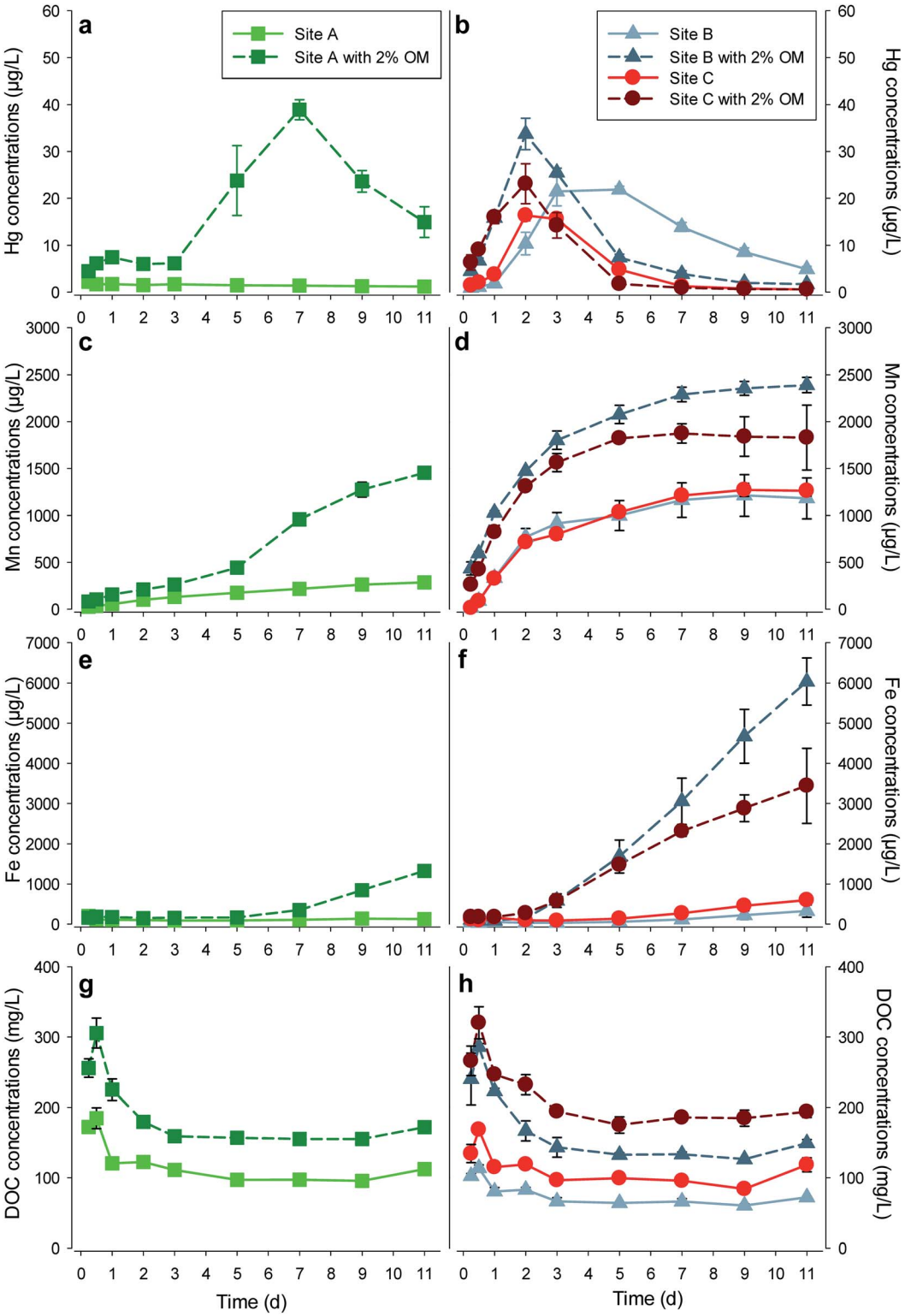


Fig. 4 Hg (a and b), Mn (c and d), Fe (e and f) and DOC (g and h) concentrations in soil solution during the incubation experiment. The left panels (a, c, e and g) represent site A. The right panels (b, d, f and h) represent sites B and C (mean ± sd, $n = 3$ separate microcosms).

It is likely that the significant increase of MeHg concentrations in the solid soil of the microcosms with 2% OM is related to their higher bioavailable OM concentration (measured as DOC concentration) and their lower Eh (Fig. 4 and S2†). DOC serves as a C and energy source for microbes and influences the microbial activity.[21] Higher microbial activity lowers the redox potential because O_2 is consumed.[75] As a consequence, the higher DOC concentrations might have caused greater Hg desorption which

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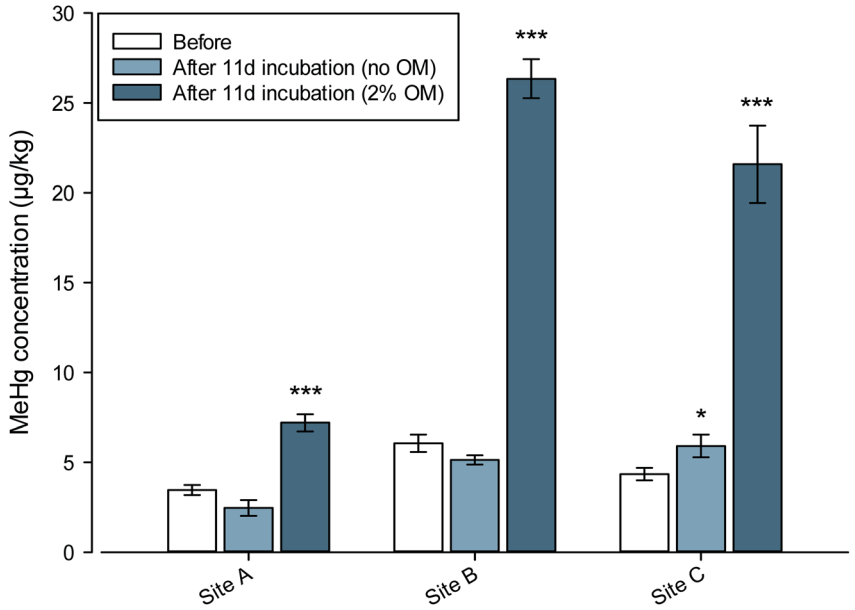


Fig. 5 MeHg concentrations in soils before incubation (white), after incubation without OM (cow dung) addition (light blue) and after incubation with 2% OM added (dark blue). Asterisks denotes a significant difference (*Z*-test) between the MeHg concentrations before and after incubation (*: $p < 0.05$, **: $p < 0.01$, ***: $p < 0.001$).

in turn increased the availability of Hg for methylation. However, Hg in the soil solution is not necessarily available for methylating bacteria because organo–Hg complexes might be too big for bacterial uptake.[9,76] It is accepted that anoxic conditions tend to favor Hg methylation, whereas oxic conditions seem to promote demethylation.[9] The different physico-chemical conditions together with possible differences in the composition of the microbial community at site A seem to have inhibited methylation or promoted demethylation compared to sites B and C, although the release of Hg in soil solution was the highest at site A (Fig. 4). The fact that MeHg concentrations in microcosms of sites B and C were similar (site B: 26.4 ± 1.1 µg kg^{-1} and site C: 21.6 ± 2.2 µg kg^{-1} Fig. 5), despite having very different soil Hg concentrations, is interesting. We suggest that Hg methylation is more related to the Hg concentration in the soil solution than in the solid soil (Fig. 4).

Different studies have reported an enhanced Hg methylation due to flooding in terrestrial ecosystems. For example, Wang *et al.*[37] found significantly lower methylation rates and a lower number of sulfur-reducing bacteria if a Hg-polluted paddy soil was kept aerobic instead of flooded in incubation experiments. This highlights that the redox milieu is a key factor controlling Hg methylation. Furthermore, the decomposing vegetation also plays a role under flooded conditions.[34–36] Commonly, ecosystem-scale studies also show higher MeHg concentrations in soils and peat after flooding. Rolfhus *et al.*[36] found a 52-fold increase of MeHg concentrations due to flooding. Further, 86% of the produced MeHg remained after 9 years of non-flooded conditions. This indicates that MeHg can be more persistent than previously thought and could explain the levels of MeHg found in the oxic soils of the three investigated fields (Fig. 3, Table 3).

Studies that investigate the effect of agricultural manure amendment (or OM addition in general) on the Hg methylation are rare. However, the similarity of our results to the study by Liu *et al.*[40] are interesting. The latter authors found in an incubation experiment a two times higher MeHg concentration if Hg-contaminated paddy soil samples were treated with rice straw. However, our MeHg concentrations were up to ten times higher and the effect within the treatments was more pronounced, possibly because animal manure introduces more readily available C and more nutrients than rice straw.

Conclusion

A selective extraction method is crucial in order to measure MeHg in soil samples in an effective way using HPLC. The modified extraction method for lipophilic organic mercury with subsequent analysis using HPLC coupled to ICP-MS presented here showed an excellent recovery of MeHg from standard reference materials at a high precision. Furthermore, the comparison with a conventional method showed a good fit of results for our samples. This method shows great potential and should be further improved to lower the LOD and increase sample throughput.

In the soils of the three studied fields situated in a polluted agricultural floodplain, the two observed gradients in Hg concentrations (with increasing soil depth and distance from the canal, from which the Hg pollution originated) confirmed the historical canal maintenance practices which resulted in the redistribution of dredged sediments to the soils next to the canal. In higher polluted soil samples (close to the canal and 0–30 cm depth) we measured MeHg concentrations between 1.2 to 7.8 µg kg^{-1}. However, in contrast to total Hg, no spatial MeHg

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gradient was found. The MeHg/Hg concentration ratio was always below 0.1%, which agrees with other studies of Hg polluted soils.[37,55,66]

We observed a strong Hg release into the soil solution within a few days during an incubation experiment with the flooded Hg-polluted soils, even in less contaminated soils such as site C. In OM (cow dung) treated microcosms, a part of the bioavailable Hg^{2+} in the soil solution was methylated during the experiment. By the end of the experiment, OM-treated soils showed a 2 to 5 times higher MeHg concentration than before the experiment. This demonstrates that upon flooding, Hg can be remobilized as well as methylated and therefore can become available to plants and soil-dwelling organisms. Further, these results show that the potential release of Hg and its subsequent methylation does not depend on the total Hg concentration in soil and that even less contaminated sites could pose a similar environmental risk as the more polluted ones. It is important to realize that common agricultural practices such as manure amendment, can provoke increased methylation rates if the soil is at least episodically saturated. More experimental studies are needed to assess the risks associated with different agricultural practices such as manure, slurry and sewage sludge amendments as well as straw return on fields, especially in Hg-polluted floodplain soils.

Conflicts of interest

There are no conflicts to declare.

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